MA2031

Linear Algebra for Engineers

Compressing Book Theory into relevant portions for the Exam, does not include problems.

Concepts before Quiz 2

A. Vector Spaces, Subspaces and Bases

Definition of a Vector Space:

A nonempty set V with two operations: $+$ (addition) that associates any two elements u, v in V to a single element $u + v$ in V, and $\cdot$ (scalar multiplication) that associates a number $\alpha \in \mathbb{F}$ and an element v in V to an element $\alpha \cdot v$ in V, is said to be a **vector space** over $\mathbb{F}$ iff it satisfies the following conditions:

(1) For all $x, y \in V$, $x + y = y + x$.

(2) For all $x, y, z \in V$, $(x + y) + z = x + (y + z)$.

(3) There exists an element $0 \in V$ such that $x + 0 = x$ for all $x \in V$.

(4) For each $x \in V$, there exists $(-x) \in V$ such that $x + (-x) = 0$.

(5) For each $x \in V$, $1 \cdot x = x$.

(6) For all $\alpha, \beta \in \mathbb{F}$ and for all $x, y \in V$, $(\alpha\beta) \cdot x = \alpha \cdot (\beta \cdot x)$.

(7) For each $\alpha \in \mathbb{F}$ and for all $x, y \in V$, $\alpha \cdot (x + y) = (\alpha \cdot x) + (\alpha \cdot y)$.

(8) For all $\alpha, \beta \in \mathbb{F}$ and for each $x \in V$, $(\alpha + \beta) \cdot x = (\alpha \cdot x) + (\beta \cdot x)$.

- If the underlying field F is R we say that V is a real vector space; and if F = C, we say that V is a complex vector space. Elements of a vector space are called vectors.
- For looking at some examples of vector spaces go to page 5-6 of Book 3

Definition of a Sub space

Let V be a vector space. A subset U of V is called a **subspace of** V iff the following conditions are satisfied:

(1) $U \neq \emptyset$.

(2) For all $x, y \in U$, $x + y \in U$.

(3) For each $x \in U$ and for each $\alpha \in \mathbb{F}$, $\alpha x \in U$.

A subspace of V which is not equal to V is called a **proper subspace** of V.

Notice that the second and the third conditions together, in the definition of a subspace, is equivalent to the following single condition:

For all $x, y \in U$ and for each $\alpha \in \mathbb{F}$, $x + \alpha y \in U$.

- In fact, a subspace is a vector space on its own right, with the operations inherited from the parent vector space.

(1.5) *Theorem*

The intersection of two subspaces of a vector space is also a subspace.

(1.6) *Theorem*

Union of two subspaces is a subspace iff one of them is a subset of the other.

Definition of Linear Combination

Let V be a vector space. Let $v_1, \ldots, v_n \in V$. A vector $v \in V$ is said to be a **linear combination** of $v_1, \ldots, v_n$ iff $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$ for some scalars $\alpha_1, \ldots, \alpha_n \in \mathbb{F}$.

Definition of Span

Let S be a nonempty subset of V. Then the set of all linear combinations of elements of S is called the **span** of S, and is denoted by span (S). Further, span of the empty set is taken to be $\{0\}$.

(1.8) *Theorem*

The span of a subset is the minimal subspace that contains the subset.

(1.9) *Theorem*

The sum of two subspaces of a vector space is equal to the span of their union.

Definition of Linear Dependence/Independence

Let S be a subset of a vector space V. S is said to be **linearly dependent** iff there exists a vector $v \in S$ such that $v \in$ span $(S \setminus \{v\})$.

A list of vectors $v_1, \ldots, v_n$ is said to be **linearly dependent** iff either a vector is repeated in the list or the set $\{v_1, \ldots, v_n\}$ is linearly dependent, or both.

A set or a list is called **linearly independent** iff it is not linearly dependent.

When a list of vectors $v_1, \ldots, v_n$ is linearly dependent (or independent), we say that the vectors $v_1, \ldots, v_n$ are linearly dependent (or independent).

Observe that $\varnothing$ is linearly independent, and $\{0\}$ is linearly dependent. Moreover, a set of at least two vectors is linearly dependent iff one of the vectors in the set is a linear combination of others.

(1.12) *Theorem*

Let $v_1, \ldots, v_n$ be vectors in a vector space V, where $n \geq 2$. Then

(1) *$\{v_1, \ldots, v_n\}$ is linearly dependent iff $\alpha_1 v_1 + \cdots + \alpha_n v_n = 0$ where at least one of the scalars $\alpha_1, \ldots, \alpha_n$ is nonzero.*

(2) *$\{v_1, \ldots, v_n\}$ is linearly independent iff for scalars $\alpha_1, \ldots, \alpha_n$,*

$$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0 \text{ implies } \alpha_1 = \cdots = \alpha_n = 0.$$

(1.15) *Theorem*

Let V be a vector space. Let A be a linearly independent set of m vectors, and let B be a spanning set of V consisting of n vectors. Then $m \leq n$.

Definition of Basis

(1.15) *Theorem*

Let V be a vector space. Let A be a linearly independent set of m vectors, and let B be a spanning set of V consisting of n vectors. Then $m \leq n$.

(1.17) *Theorem*

Any subset of a vector space is a basis iff it is a minimal spanning set iff it is a maximal linearly independent set.

(1.18) *Theorem*

Let $v_1, \ldots, v_n$ be vectors in a vector space V. The ordered set $B = \{v_1, \ldots, v_n\}$ is a basis of V iff for each $v \in V$ there exists a unique n-tuple of scalars $(\alpha_1, \ldots, \alpha_n)$ such that $v = \alpha_1 v_1 + \cdots + \alpha_n v_n$.

(1.19) *Theorem*

If a vector space has a finite spanning set, then every finite spanning set contains a basis.

(1.21) *Theorem*

If a vector space has a finite spanning set, then every linearly independent set can be extended to a basis.

Definition of a dimension

(1.23) *Theorem*

If a vector space has a finite spanning set, then each basis has the same number of elements.

Let V be a vector space having a finite spanning set. Then the number of elements in a basis is called the **dimension** of V; and it is denoted by $\dim(V)$.

(1.29) ***Theorem***

Let U and W be finite dimensional subspaces of a vector space V. Then

$$\dim(U \cap W) + \dim(U + W) = \dim(U) + \dim(W).$$

Extracting a Basis

A question of computational importance is that given a finite list of vectors, how do we construct a basis for the subspace spanned by these vectors?

The notion of elementary operations help in achieving this, at least in $\mathbb{F}^n$. We observe the following connections between linear combinations, linear dependence and linear independence of rows and columns of a matrix, and its row reduced echelon form (RREF).

For example,

Let $U = \text{span}\{v_1, v_2, v_3, v_4, v_5\}$ in $\mathbb{R}^4$, where $v_1 = (1, 1, 0, -1)$, $v_2 = (2, -1, 1, 0)$, $v_3 = (1, 2, -2, 1)$, $v_4 = (1, 5, -3, -1)$, $v_5 = (4, -1, 0, 2)$. It is required to extract a basis for U from the list of these vectors.

Method 1: Taking the vectors as rows of a matrix and reducing it to its RREF, we have

$$\begin{bmatrix} 1 & 1 & 0 & -1 \\ 2 & -1 & 1 & 0 \\ 1 & 2 & -2 & 1 \\ 1 & 5 & -3 & -1 \\ 4 & -1 & 0 & 2 \end{bmatrix} \xrightarrow{RREF} \begin{bmatrix} 1 & 0 & 0 & 1/5 \\ 0 & 1 & 0 & -6/5 \\ 0 & 0 & 1 & -8/5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Discarding the zero rows, we obtain the basis for U as

$$\{(1, 0, 0, 1/5),\ (0, 1, 0, -6/5),\ (0, 0, 1, -8/5).$$

Method 2: Taking the vectors as columns of a matrix and reducing it to its RREF, we have

$$\begin{bmatrix} 1 & 2 & 1 & 1 & 4 \\ 1 & -1 & 2 & 5 & 1 \\ 0 & 1 & -2 & -3 & 0 \\ -1 & 0 & 1 & -1 & 2 \end{bmatrix} \xrightarrow{RREF} \begin{bmatrix} 1 & 0 & 0 & 2 & -1 \\ & 1 & 0 & -1 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

In the RREF, the first three columns are pivoted columns and the last two are non-pivoted. Therefore, a basis for U is $\{v_1, v_2, v_3\}$. Further, the entries in the non-pivoted columns show that $v_4 = 2v_1 - v_2 + v_3$ and $v_5 = -v_1 + 2v_2 + v_3$. □

Quotient Spaces – Have we studied this in class?

B. Linear Transformation, Matrix Representation, Change of Basis, Rank Nullity Theorem

Definition of Linear Transformation/ Linear Map

The interesting maps in a domain of mathematical discourse are those maps which preserve structures. Since a vector space gets its structure from the two operations of addition and scalar multiplication, we are interested in maps that preserve these operations.

Let V and W be vector spaces over the same field $\mathbb{F}$. A function $T : V \to W$ is said to be a **linear transformation** (or a linear map, or a linear operator) iff for all $x, y \in V$ and for each $\alpha \in \mathbb{F}$,

$$T(x + y) = T(x) + T(y) \text{ and } T(\alpha x) = \alpha T(x).$$

A linear transformation from V to V is called a **linear operator on** V.

A linear transformation from V to $\mathbb{F}$ is called a **linear functional**.

Observe that the two conditions in the definition of a linear transformation amount to the single condition

$$T(x + \alpha y) = T(x) + \alpha T(y) \quad \text{for all } x, y \in V, \text{ and for each } \alpha \in \mathbb{F}.$$

- **Examples of many linear maps** are given on 52-53 of Book 3, these must be looked into

Theorems related to linear transformations or maps

(3.2) *Theorem*

Let $T : V \to W$ be a linear transformation. Then the following are true:

(1) $T(0) = 0$.

(2) *For all* $u, v \in V$, $T(u - v) = T(u) - T(v)$.

(3) *For any* $n \in \mathbb{N}$, *for all* $v_1, \ldots, v_n \in V$ *and for all scalars* $\alpha_1, \ldots, \alpha_n$, $T(\alpha_1 v_1 + \cdots + \alpha_n v_n) = \alpha_1 T(v_1) + \cdots + \alpha_n T(v_n)$.

(3.3) *Theorem*

Let $T : U \to V$ and $S : V \to W$ be linear transformations. Then $S \circ T : U \to W$ is a linear transformation.

(3.4) *Theorem*

A linear transformation is uniquely determined from its action on a basis.

Definition of Null Space and the Range Space of a Linear Transformation

Let $T : V \to W$ be a linear transformation. Then

$N(T) = \{v \in V : T(v) = 0\}$ is called the **null space** of T;

$R(T) = \{T(v) : v \in V\}$ is called the **range space** of T.

The null space of T is also called the *kernel* of T. We should justify as to why these two sets of vectors are called *spaces.*

(3.6) *Theorem*

Let $T : V \to W$ be a linear transformation. Then $N(T)$ is a subspace of V and $R(T)$ is a subspace of W.

(3.8) *Theorem*

Let $T : V \to W$ be a linear transformation. Let B be a basis of V. Then

$$R(T) = \text{span}\,\{T(v) : v \in B\}.$$

Using Nullity and Rank to determine whether the map is one-one or onto

(3.9) *Theorem*

Let $T : V \to W$ be a linear transformation.

(1) *T is one-one iff $N(T) \subseteq \{0\}$ iff $N(T) = \{0\}$ iff* $\text{null}(T) = 0$.

(2) *T is an onto map iff $W \subseteq \text{span}\,\{Tv : v \in B\}$ for any basis B of V iff* $\text{rank}(T) = \dim(W)$.

Rank Nullity Theorem

Let V be a finite dimensional vector space. Let $T : V \to W$ be a linear transformation. Then $\text{rank}(T) + \text{null}(T) = \dim(V)$.

Definition of Isomorphism

A one-one and onto linear map is called an isomorphism

Two vector spaces are called **isomorphic** to each other iff there exists an isomorphism from one to the other.

(3.13) *Theorem*

The inverse of an isomorphism is also an isomorphism.

(3.14) *Theorem*

Let V and W be finite dimensional vector spaces with $\dim(V) = \dim(W)$. Let $T : V \to W$ be a linear transformation. Then T is an isomorphism iff T is one-one iff T it is onto.

(3.16) *Theorem*

A vector space is isomorphic to a finite dimensional vector space iff both the spaces have equal dimensions.

(3.17) *Theorem*

Let $P : U \to V$, $T : V \to W$ and $Q : W \to X$ be linear transformations, where V and W are finite dimensional vector spaces. Suppose P and Q are isomorphisms. Then $\text{rank}(QTP) = \text{rank}(T)$ and $\text{null}(QTP) = \text{null}(T)$.

(3.18) *Theorem*

Let U, V, W and X be finite dimensional vector spaces with $\dim(U) = \dim(V)$ and $\dim(W) = \dim(X)$. Let $T : V \to W$ and $S : U \to X$ be linear transformations. If $\text{rank}(S) = \text{rank}(T)$, then there exist isomorphisms $P : U \to V$ and $Q : W \to X$ such that $S = QTP$.

Matrix of a Linear Transformation

(4.15) *Theorem*

Each coordinate vector map is an isomorphism.

That is, the coordinate vector of $T(v)$ can be obtained once we know the coordinate vectors of $T(v_1), \ldots, T(v_n)$ with respect to E. Suppose the coordinate vectors of $T(v_j)$ are given as follows:

$$[T(v_1)]_E = \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix}, \cdots, [T(v_j)]_E = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{bmatrix}, \cdots, [T(v_n)]_E = \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

for scalars a_{ij}. Notice that this is equivalent to expressing $T(v_j)$ in terms of the basis vectors $w_1, \ldots, w_m$ as in the following:

$$\begin{aligned} T(v_1) &= a_{11}w_1 + \cdots + a_{m1}w_m \\ &\vdots \\ T(v_n) &= a_{1n}w_1 + \cdots + a_{mn}w_m. \end{aligned}$$

We put together the coordinate vectors $[T(v_j)]_E$ in that order to obtain the following array of scalars a_{ij} :

$$\left[[T(v_1)]_E \cdots [T(v_j)]_E \cdots [T(v_n)]_E \right] = \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ & & \vdots & & \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ & & \vdots & & \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}.$$

Definition of Matrix of linear transformation

We summarize and give a notation. Let V and W be vector spaces over $\mathbb{F}$ with dimensions n and m, and ordered bases $B = \{v_1, \ldots v_n\}$ and $E = \{w_1, \ldots, w_m\}$, respectively. Let $T : V \to W$ be a linear transformation. Let $[T(v_j)]_E$ denote the coordinate vector of $T(v_j)$ with respect to E. The **matrix** $[T]_{E,B}$ **of** T **with respect to the ordered bases** B **and** E is given by

$$[T]_{E,B} := \Big[[T(v_1)]_E \ \ldots \ [T(v_n)]_E \Big].$$

(4.17) ***Theorem***

Let $B = \{v_1, \ldots, v_n\}$ and $E = \{w_1, \ldots, w_m\}$ be ordered bases for the vector spaces V and W, respectively. Let $T : V \to W$ be a linear transformation. Then $[T]_{E,B}$ is the unique matrix in $\mathbb{F}^{m \times n}$ such that for each $x \in V$,

$$[Tx]_E = [T]_{E,B}[x]_B.$$

(4.18) *Theorem*

Let B and E be ordered bases for the finite dimensional vector spaces V and W, respectively. Let $S, T : V \to W$ be linear transformations. Let $\alpha \in \mathbb{F}$. Then

$$[S+T]_{E,B} = [S]_{E,B} + [T]_{E,B} \quad \text{and} \quad [\alpha T]_{E,B} = \alpha [T]_{E,B}.$$

(4.19) *Theorem*

Let B, C and D be ordered bases for the finite dimensional vector spaces U, V and W, respectively. Let $S : U \to V$ and $T : V \to W$ be linear transformations. Then

$$[TS]_{D,B} = [T]_{D,C}\,[S]_{C,B}.$$

(4.20) *Theorem*

Let B and E be ordered bases for finite dimensional vector spaces V and W, respectively. Let $T : V \to W$ be an isomorphism. Then

$$[T^{-1}]_{B,E} = \left([T]_{E,B}\right)^{-1}.$$

(4.21) *Theorem*

Let $B = \{v_1, \ldots, v_n\}$ and $E = \{w_1, \ldots, w_m\}$ be ordered bases of the inner product spaces V and W, respectively. Let $T : V \to W$ be a linear transformation. If E is an orthonormal basis of W, then the (ij)th entry of $[T]_{E,B}$ is equal to $\langle Tv_j, w_i \rangle$.

(4.22) *Theorem*

Let $B = \{v_1, \ldots, v_n\}$ and $E = \{w_1, \ldots, w_m\}$ be orthonormal ordered bases of the ips V and W, respectively. Let $T : V \to W$ be a linear transformation. Then $[T^*]_{B,E} = ([T]_{E,B})^*$.

(4.24) *Theorem*

Let T be a linear operator on a finite dimensional inner product space V. Let B be an orthonormal ordered basis of V.

(1) *T is self-adjoint iff $[T]_{B,B}$ is hermitian.*

(2) *T is normal iff $[T]_{B,B}$ is normal.*

(3) *T is unitary iff $[T]_{B,B}$ is unitary.*

Change of Basis

Suppose V and W are vector spaces of dimensions n and m, respectively. If we fix ordered bases for V and W, then the coordinate vector maps with respect to these bases provide isomorphisms from V and W to $\mathbb{F}^{n\times 1}$ and $\mathbb{F}^{m\times 1}$. Using such coordinate vector maps, we had seen how to represent a linear transformation from V to W by a matrix.

To fix the ideas, let $B = \{v_1, \ldots, v_n\}$ be an ordered basis of V. Let $C = \{w_1, \ldots, w_m\}$ be an ordered basis of W. Have standard bases for $\mathbb{F}^{n\times 1}$ and $\mathbb{F}^{m\times 1}$. The coordinate vector maps $[\ ]_B : V \to \mathbb{F}^{n\times 1}$ and $[\ ]_C : W \to \mathbb{F}^{m\times 1}$ are isomorphisms. Let $T : V \to W$ be a linear transformation and $[T]_{C,B}$ be its matrix representation. Then we have the following commutative diagram:

$$\begin{array}{ccc} V & \xrightarrow{\ T\ } & W \\ {\scriptstyle [\ ]_B}\downarrow{\scriptstyle \simeq} & & {\scriptstyle \simeq}\downarrow{\scriptstyle [\ ]_C} \\ \mathbb{F}^{n\times 1} & \xrightarrow[[T]_{C,B}]{} & \mathbb{F}^{m\times 1} \end{array}$$

It means

$$T = [\ ]_C^{-1} \circ [T]_{C,B} \circ [\ ]_B, \quad [T]_{C,B} = [\ ]_C \circ T \circ [\ ]_B^{-1}. \tag{4.5.1}$$

Also, $[\ ]_C \circ T = [T]_{C,B} \circ [\ ]_B$, which amounts to

$$[T(x)]_C = [T]_{C,B}\,[x]_B \quad \text{for each } x \in V. \tag{4.5.2}$$

We know that isomorphisms preserve rank and nullity. Since the coordinate vector maps are isomorphisms, we obtain the following theorem.

(4.25) *Theorem*

Let V and W be finite dimensional vector spaces with ordered bases B and C, respectively. Let $T : V \to W$ be a linear transformation. Then

$$\text{rank}(T) = \text{rank}([T]_{C,B}) \quad \textit{and} \quad \text{null}(T) = \text{null}([T]_{C,B}).$$

(4.28) *Theorem*

Let $A \in \mathbb{F}^{m\times n}$. Let $B = \{v_1, \ldots, v_n\}$ and $C = \{w_1, \ldots, w_m\}$ be ordered bases for $\mathbb{F}^{n\times 1}$ and $\mathbb{F}^{m\times 1}$, respectively. Then $[A]_{C,B} = Q^{-1}AP$, where

$$Q = [w_1 \ \cdots \ w_m] \quad \textit{and} \quad P = [v_1 \ \cdots \ v_n].$$

Equivalence of 2 Matrices

Let $A, B \in \mathbb{F}^{m\times n}$. We say that B is **equivalent to** A iff there exist invertible matrices $P \in \mathbb{F}^{n\times n}$ and $Q \in \mathbb{F}^{m\times m}$ such that $B = Q^{-1}AP$.

As it happens, a change of bases in both the domain and co-domain spaces bring up an equivalent matrix that represents the original matrix viewed as a linear transformation.

It is easy to see that on $\mathbb{F}^{m\times n}$, the relation 'is equivalent to' is an equivalence relation. Observe that A and B are equivalent iff there exist invertible matrices P and Q of appropriate order such that $B = QAP$. There is an easy characterization of equivalence of two matrices.

C. Inner Products

Definition

An **inner product** on a vector space V over $\mathbb{F}$ is a map from $V \times V$ to $\mathbb{F}$, which associates a pair of vectors $x, y \in V$ to a scalar $\langle x, y\rangle$ in $\mathbb{F}$ satisfying the following properties:

(1) For each $x \in V$, $\langle x, x\rangle \geq 0$.

(2) For each $x \in V$, $\langle x, x\rangle = 0$ iff $x = 0$.

(3) For all $x, y, z \in V$, $\langle x + y, z\rangle = \langle x, z\rangle + \langle y, z\rangle$.

(4) For each $\alpha \in \mathbb{F}$ and for all $x, y \in V$, $\langle \alpha x, y\rangle = \alpha\langle x, y\rangle$.

(5) For all $x, y \in V$, $\langle y, x\rangle = \overline{\langle x, y\rangle}$.

The usual dot product on $\mathbb{R}^2$ is an inner product on $\mathbb{R}^2$.

- **Several examples of Inner products given on pages 37-38 of Book 3**

Definition of the norm of a vector

Let V be an ips. For any $x \in V$, the length of x, also called the **norm** of x is defined as

$$\|x\| = \sqrt{\langle x, x\rangle}.$$

Notice that $\langle x, x\rangle \geq 0$ implies that $\|x\|$ is a real number.

1. For each $x \in V$, $\|x\| \geq 0$.
2. For each $x \in V$, $x = 0$ iff $\|x\| = 0$.
3. For each $x \in V$ and for each $\alpha \in \mathbb{F}$, $\|\alpha x\| = |\alpha|\|x\|$.

Some properties of inner products and associated quantities like the norm

For all vectors x, y in an ips, the following are true:

(1) **(Parallelogram Law)** $\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$.

(2) **(Cauchy-Schwartz Inequality)** $|\langle x, y\rangle| \leq \|x\|\,\|y\|$.
Further, equality holds iff one of x, y is a scalar multiple of the other.

(3) **(Triangle Inequality)** $\|x+y\| \leq \|x\| + \|y\|$.

(4) **(Reverse Triangle Inequality)** $\big|\,\|x\| - \|y\|\,\big| \leq \|x-y\|$.

Definition of the Angle between x and y

$$\cos\theta(x,y) = \frac{|\langle x, y\rangle|}{\|x\|\,\|y\|}.$$

Let x, y be vectors in an ips. We say that x is **orthogonal** to y iff $\langle x, y\rangle = 0$. When x is orthogonal to y, we write $x \perp y$.

Thus the zero vector is orthogonal to every vector, and $x \perp y$ implies $y \perp x$.

(2.5) ***Theorem*** *Pythagoras*

Let x and y be vectors in an ips V.

(1) *If $x \perp y$, then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$.*

(2) *If V is a real ips, then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$ implies $x \perp y$.*

Relations between orthogonality, inner products and linear independence

(2.7) ***Theorem***

Every orthogonal (orthonormal) set is linearly independent.

(2.8) ***Theorem***

Let $B = \{v_1, \ldots, v_n\}$ be an orthonormal basis of an ips V. Let $x \in V$. Then

(1) **(Fourier Expansion)** $x = \sum_{j=1}^{n} \langle x, v_j\rangle v_j$;

(2) **(Parseval's Identity)** $\|x\|^2 = \sum_{j=1}^{n} |\langle x, v_j\rangle|^2$.

(2.9) ***Theorem*** (Bessel's Inequality)

Let $E = \{u_1, \ldots, u_m\}$ be an orthonormal set in an ips V. Let $x \in V$. Then $\sum_{j=1}^{m} |\langle x, u_j\rangle|^2 \leq \|x\|^2$.

How to convert a list of dependent vectors to linearly independent vectors?

Gram Schimdt Orthogonalisation

(2.10) *Theorem*

Let $u_1, \ldots, u_n$ be linearly independent vectors in an ips V. Construct the vectors $v_1, \ldots, v_n$ as follows:

$$v_1 = u_1$$
$$v_k = u_k - \frac{\langle u_k, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 - \frac{\langle u_k, v_2 \rangle}{\langle v_2, v_2 \rangle} v_2 - \cdots - \frac{\langle u_k, v_{k-1} \rangle}{\langle v_{k-1}, v_{k-1} \rangle} v_{k-1} \quad \text{for } k > 1.$$

Then span $\{v_1, \ldots, v_n\}$ = span $\{u_1, \ldots, u_n\}$, *and* $\{v_1, \ldots, v_n\}$ *is orthogonal.*

(2.13) *Theorem*

An orthonormal set in a finite dimensional ips can be extended to an orthonormal basis. Every finite dimensional ips has an orthonormal basis.

Concepts after Quiz 2

A. Adjoint Linear Transformation:

Prerequisite theorem:

Let V and W be inner product spaces, where V is of finite dimension. Then, corresponding to each linear transformation $T : V \to W$, there exists a unique linear transformation $S : W \to V$ such that $\langle Tv, w \rangle = \langle v, Sw \rangle$ for all $v \in V$, $w \in W$.

This theorem hints so as to how a unique linear transformation corresponding to every linear transformation can be found. This linear transformation will be called as the adjoint of the previous linear transformation.

Definition of the Adjoint: T^*

$$\langle Tx, y \rangle = \langle x, T^* y \rangle \quad \text{for all } x \in V,\ y \in W$$

How to calculate the adjoint?

If $\{v_1, \ldots, v_n\}$ is an orthonormal
basis of V, and $T : V \to W$ is a linear transformation, then $T^* : W \to V$ is given by

$$T^*(w) = \sum_{i=1}^{n} \langle w, T(v_i) \rangle v_i \quad \text{for each } w \in W.$$

Riesz Representation:

Prerequisite theorem for this representation:

Corresponding to each linear functional f on a finite dimensional ips V, there exists a unique $y \in V$ such that for each $x \in V$, $f(x) = \langle x, y \rangle$.

What is a linear functional?

It is basically a linear transformation that has the domain as the vector space (or the ips) and the codomain as the field over which the vector space is defined. For example, cartesian dot product is a linear functional.

Riesz Representer

Call the vector y in the equation $f(x) = \langle x, y \rangle$ as the **Riesz representer**

Calculating the Riesz Representer

$$v_f = \sum_{i=1}^{n} \overline{f(v_i)}\, v_i.$$

if $\{v_1, \ldots, v_n\}$ is an orthonormal basis of V

The Riesz representer can be used as an independent concept for showing the existence of a unique adjoint of a linear transformation.

Additional Relations and Formulae related to Adjoint:

Let U, V and W be finite dimensional ips. Let $S : U \to V$ and $T, T_1, T_2 : V \to W$ be linear transformations. Let $I : V \to V$ be the identity operator and let $\alpha \in \mathbb{F}$. Then

$$(T_1 + T_2)^* = T_1^* + T_2^*, \quad (\alpha T)^* = \overline{\alpha}\, T^*, \quad (T^*)^* = T, \quad I^* = I, \quad (TS)^* = S^* T^*.$$

Let V and W be finite dimensional inner product spaces, and let $T : V \to W$ be a linear transformation. Then the following are true:

(1) $N(T^*T) = N(T)$, $N(TT^*) = N(T^*)$.

(2) $\text{rank}(T^*) = \text{rank}(T^*T) = \text{rank}(TT^*) = \text{rank}(T)$.

(3) $R(T^*T) = R(T^*)$, $R(TT^*) = R(T)$.

(4) *If* $\dim(V) = \dim(W)$, *then* $\text{null}(T^*) = \text{null}(T^*T) = \text{null}(TT^*) = \text{null}(T)$.

Basing on the adjoint, special types of linear operators can be defined.
A linear operator T on a finite dimensional ips V is called

1. **self-adjoint** iff $T^* = T$;
2. **normal** iff $T^*T = TT^*$;
3. **unitary** iff $T^*T = TT^* = I$;
4. **orthogonal** iff $T^*T = TT^* = I$ and V is a real ips;
5. **isometric** iff $\|Tx\| = \|x\|$ for each $x \in V$.

Each self-adjoint linear operator is normal, and each unitary linear operator is normal and invertible. Similarly, it can be shown that a linear operator on a finite dimensional ips is unitary iff it is isometric

B. Orthogonality

There are 3 concepts that come under this category : QR Factorisation, Best Approximation and Least Squares Problem

1. QR Factorisation:

Prerequisite Theorem:

Theorem 4.3 (QR-factorization) *Let the columns of $A \in \mathbb{F}^{m\times n}$ be linearly independent. Then there exist a matrix $Q \in \mathbb{F}^{m\times n}$ with orthonormal columns, and an upper triangular invertible matrix $R \in \mathbb{F}^{n\times n}$ such that $A = QR$. Further, if the diagonal entries in R are real, and they are either all positive, or all negative, then both Q and R are uniquely determined from A.*

How to find Q and R?

For an orthonormal list of vectors $w_1, w_2, \ldots, w_n$

We can use gram schimdt orthonormalization to find such a factorisation

Finding R

$$
\begin{aligned}
u_1 &= a_{11}w_1 \\
u_2 &= a_{12}w_1 + a_{22}w_2 \\
&\vdots \\
u_k &= a_{1k}w_1 + \cdots + a_{kk}w_k \\
&\vdots \\
u_n &= a_{1n}w_1 + \cdots + a_{kn}w_k + \cdots + a_{nn}w_n
\end{aligned}
$$

We take $a_{ij} = 0$ for $i > j$, and form the matrix

$$
R = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ & a_{22} & \cdots & a_{2n} \\ & & \ddots & \vdots \\ & & & a_{nn} \end{bmatrix}.
$$

Finding Q

Construct $Q = [w_1 \quad \cdots \quad w_n]$.

This factorisation is unique.

R can also be found using this equation

$$R = Q^* A.$$

Orthogonal Projection

Important Identities

Theorem 4.4 *Let $\{v_1, \ldots, v_m\}$ be an orthonormal basis for a subspace V of $\mathbb{F}^n$. Then the following are true for each $x \in V$:*

(1) **(Fourier Expansion)** $x = \langle x, v_1\rangle v_1 + \cdots + \langle x, v_m\rangle v_m$.
(2) **(Parseval Identity)** $\|x\|^2 = |\langle x, v_1\rangle|^2 + \cdots + |\langle x, v_m\rangle|^2$.

Prerequisite Theorem for the existence of the orthogonal projection

Theorem 4.5 *Let V be a subspace of $\mathbb{F}^n$. Corresponding to each $x \in \mathbb{F}^n$, there exists a unique vector $y \in V$ such that $x - y \perp V$. Further, if $\{v_1, \ldots, v_m\}$ is any orthonormal basis of V, then such a vector y is given by $y = \sum_{j=1}^{m} \langle x, v_j \rangle v_j$.*

Notation for such a y: $\text{proj}_V(x)$.

Bessel's Inequality:

Theorem 4.6 (Bessel Inequality) *Let $\{v_1, \ldots, v_m\}$ be an orthonormal basis for a subspace V of $\mathbb{F}^n$. Then for each $x \in \mathbb{F}^n$, $\|x\|^2 \geq \|\text{proj}_V(x)\| = \sum_{j=1}^{m} |\langle x, v_j \rangle|^2$.*

The concept of calculating the projection can be seen as a matrix product as well:

For this, let $\{v_1, \ldots, v_m\}$ be an orthonormal basis of a subspace V of $\mathbb{F}^{n\times 1}$. Let $x \in \mathbb{F}^{n\times 1}$. Write $z = [c_1, \cdots, c_m]^t \in \mathbb{F}^{m\times 1}$ with $c_j = \langle x, v_j \rangle = v_j^* x$, and $P = [v_1 \quad \cdots \quad v_m]$. Then

$$z = \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix} = \begin{bmatrix} v_1^* x \\ \vdots \\ v_m^* x \end{bmatrix} = P^* x, \quad y = \sum_{j=1}^{m} \langle x, v_j \rangle v_j = \sum_{j=1}^{m} c_j v_j = Pz = PP^* x.$$

Notice that $PP^* x = \text{proj}_V(x)$ for each $x \in \mathbb{F}^{n\times 1}$. Due to this reason, the matrix $PP^* \in \mathbb{F}^{n\times n}$ is called the **projection matrix** that projects vectors of $\mathbb{F}^{n\times 1}$ onto the subspace V.

2. Best Approximation and Least Squares Solution

Concept behind Best Approximation:

Let V be a subspace of $\mathbb{F}^n$ and let $x \in \mathbb{F}^n$. A vector $u \in V$ is called a **best approximation** of x from V iff $\|x - u\| \leq \|x - v\|$ for each $v \in V$.

Linking the Orthogonal Projection and the Best Approximation:

Theorem 4.7 *Let V be a subspace of $\mathbb{F}^n$ and let $x \in \mathbb{F}^n$. Then $\text{proj}_V(x)$ is the unique best approximation of x from V.*

Now due to the previous results, there are now multiple ways to calculate the best approximation of x from V.

Either computing projV (x) by employing an orthonormal basis, or by using the orthogonality condition directly

First Approach:

Starting from any basis for V, we employ Gram–Schmidt orthonormalization to obtain an orthonormal basis $\{v_1, \ldots, v_m\}$ for V. Then the best approximation y of x from V is given by

$$y = \langle x, v_1\rangle v_1 + \cdots + \langle x, v_m\rangle v_m.$$

Also, the best approximation may be computed by using the projection matrix.

Second Approach:

In the second approach, we look for a vector y that satisfies the orthogonality condition:

$$x - y \perp v \quad \text{for each vector } v \in V.$$

If $\{v_1, \ldots, v_m\}$ is a basis for V, then this condition is equivalent to

$$x - y \perp v_j \quad \text{for each } j = 1, \ldots, m.$$

Since $y \in V$, we may write $y = \sum_{j=1}^{m} \alpha_j v_j$. Then we determine the scalars α_j so that for $i = 1, \ldots, m$, $\langle x - \sum_{j=1}^{m} \alpha_j v_j, v_i\rangle = 0$. That is, we solve the following linear system:

$$\begin{aligned} \langle v_1, v_1\rangle \alpha_1 + \cdots + \langle v_m, v_1\rangle \alpha_m &= \langle x, v_1\rangle \\ &\vdots \\ \langle v_1, v_m\rangle \alpha_1 + \cdots + \langle v_m, v_m\rangle \alpha_m &= \langle x, v_m\rangle \end{aligned}$$

Least Squares Solution

Definition:

Let $A \in \mathbb{F}^{m\times n}$ and let $b \in \mathbb{F}^{m\times 1}$. A vector $u \in \mathbb{F}^{n\times 1}$ is a called a **least squares solution** of the linear system $Ax = b$ iff $\|Au - b\| \leq \|Az - b\|$ for all $z \in \mathbb{F}^{m\times 1}$.

Important Results related to the Least Squares Solution

Theorem 4.8 *Let $A \in \mathbb{F}^{m\times n}$ and let $b \in \mathbb{F}^{m\times 1}$.*

(1) The linear system $Ax = b$ has a least squares solution.
(2) A vector $u \in \mathbb{F}^{n\times 1}$ is a least squares solution of $Ax = b$ iff $Au - b \perp v$ for each $v \in R(A)$.
(3) A vector $u \in \mathbb{F}^{n\times 1}$ is a least squares solution of $Ax = b$ iff it is a solution of $A^*Ax = A^*b$.
(4) A least squares solution is unique iff $N(A) = \{0\}$.

C. Eigenvalues and Eigenvectors

Definition:

Let $A \in \mathbb{F}^{n\times n}$. A scalar $\lambda \in \mathbb{F}$ is called an **eigenvalue** of A iff there exists a nonzero vector $v \in \mathbb{F}^{n\times 1}$ such that $Av = \lambda v$. Such a nonzero vector v is called an **eigenvector of A for** (associated with, corresponding to) the eigenvalue λ.

The Characteristic Polynomial

Helps in computing the eigenvalues of a Matrix.

Prerequisite Theorem

Theorem 5.1 *Let $A \in \mathbb{F}^{n\times n}$. A scalar $\lambda \in \mathbb{F}$ is an eigenvalue of A iff* $\det(A - \lambda I) = 0$.

Each eigenvalue of A is a zero of the characteristic polynomial of A. Conversely, each zero of the characteristic polynomial is said to be a ***complex eigenvalue*** of A.

- The fundamental theorem of algebra states that any polynomial of degree n, with complex coefficients, can be written as a product of n linear factors each with complex coefficients. Thus, χ A(t) has exactly n, not necessarily distinct, zeros in C. And these are the eigenvalues (complex eigenvalues) of the matrix A

- If A is a matrix with real entries, some of the zeros of χ A(t) may turn out to be complex numbers with nonzero imaginary parts. Considering A as a linear transformation on Rn×1,the scalars are now real numbers. Thus each zero of the characteristic polynomial may not be an eigenvalue; only the real zeros are.

Monic Polynomial: A polynomial with the coefficient of the highest degree as 1 is called a monic polynomial.